

Bangla Image Captioning With Bidirectional GRU & Attention Mechanism

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Abstract— Image captioning is a complex study subject in which natural language subtitles/captions are generated for pictures relying upon what is shown in them. The captioning models have demonstrated impressive results in recent times and models with attention mechanisms are becoming more popular as a result of improved performance. However, the use of attention strategies for Bengali captioning challenges has received little focus. Since no study on employing attention approaches using the Bangladeshi dataset has been done yet, this paper aims at building an attention-based captioning model for the Bengali language and Bangladesh's geographic regions. With a caption-model composed of a GRU (gated recurrent network) and Bahdanau-attention, we used four CNN architectures: VGG19, InceptionV3, Xception, and ResNet50, for accurate features. The models were prepared by a public dataset named "BanglaLekhaImageCaptions" incorporating 9,154 pictures with Bangladeshi geological and cultural settings and Bengali annotations. Both qualitative and quantitative evaluations were used to assess the models. Experimental findings reveal that the models beat the existing works' results and that when the encoder is Xception, the accuracy is adequate with a BLEU-1 score of 87.18.

Keywords—Bengali Image Captioning ,CNN, GRU, Attention Mechanism, Bengali dataset.

I. INTRODUCTION

Image captioning is a process in which a machine or model of artificial intelligence learns to explain an image in human-readable language. It's a computer vision, NLP, and machine learning-related task. Image captioning can be used for things like content-based image retrieval (CBIR), online picture exploration, human-computer interaction, intelligent guiding for visually impaired people, surveillance and security, and social media image description[1]. Human minds have a remarkable ability to comprehend visual information. An image's content can be easily described by a human. It is, nevertheless, a difficult task for machine learning models. The models must be able to recognize the objects in the scene and convey the relationships between them in natural language. Furthermore, real-world scenes can be interpreted in a variety of ways.

Traditional encoder-decoder language models have the disadvantage of considering all picture information to be included in a single feature vector. This problem is solved by attention-based models. Attention processes focus on small

areas of the image, resulting in a considerable reduction in feature size. Rather than perceiving the complete image, models with attention mechanisms concentrate on a tiny portion of it and develop a name for it. During the procedure, the caption changes dynamically. These algorithms have gained a reputation for producing effective and accurate image captioning[1-6].

Image captioning has attracted the interest of researchers all across the world in recent years. There is already an extensive study on image captioning in English. Researchers have begun to use state-of-the-art captioning models for worldwide languages such as Hindi[7], Myanmar language[8], Arabic [9], Chinese [10], and Japanese [11]. Bengali is the world's fifth most widely spoken oral communication language. This language is spoken by 166 million people in Bangladesh. As a result, captioning photos in Bengali is of interest to both Bangladeshis and the rest of the globe. In recent years, we've seen a few young Bangladeshi researchers investigate image captioning in the Bengali language, with promising results [12-19]. However, there isn't enough work on Bengali attention-based captioning models. By merging attention mechanisms with encoder-decoder models, the results of Bengali captioning models can be improved. As a result, we built an attention-based model for Bengali captioning in this study, which is a more competitive model for the Bengali image captioning problem. The main purpose of this study is to develop four models for producing Bengali captions from real scenes (pictures), using four encoders (VGG19, InceptionV3, Xception, and ResNet50) for feature extraction and a bidirectional GRU with a local attention mechanism as a decoder and examine the results. When compared to state-of-the-art techniques, the models perform much better. Because it outperformed models using alternative CNNs, we chose the attention-model with Xception as our suggested model.

We arranged the rest of this paper into the sections below: The following section (Section II) covers this research's related prior studies. Section III. explains how the data in this study were prepared before the models were trained. The next part (Section IV) focuses on the specifics of how the CNNs were used to build features. Section V. provides an overview of the attention mechanism as well as how it operates. The method of our model is specified in Section VI. In section VII, we gave the models' training-related details and our paper's findings followed by some

comparisons and illustrations. Finally, in section VIII, a conclusion is added.

II. RELATED WORK

The existing captioning approaches can be divided into two categories: (1) classical image captioning: template-based and retrieval-based image captioning; and (2) deep learning-based image captioning: encoder-decoder models and attention methods. Deep learning models have made significant progress and have demonstrated that they can handle the task significantly better than traditional methods. This section focuses mostly on newly released models that have been specifically trained in Bengali.

The first comprehensive work on Bengali captioning was released in 2018 with "Chittron" by Rahman *et al.* [12]. The model was an encoder-decoder with stacked LSTM layers and a VGG16 model. They gathered 16,000 photos of Bangladeshi cultural and geographical backgrounds and personally annotated them. They evaluated the model using both quantitative and qualitative techniques. The model produced some good Bengali captions for images with low BLEU scores, according to the results.

Using the "BanglaLekhaImageCaptions"[20] dataset, the authors, Kamal *et al.* proposed a model that followed the same model architecture as "Chittron" in their paper [13]. The BLEU scores were employed for evaluation, and the results were very comparable. Meanwhile, Jishan *et al.* [14] suggested a hybrid model in which they used VGG16 as an encoder and a model consisting of B-RNN and RNN as a decoder. They improved the findings by training the hybrid model on their newly developed dataset BNLT [21] with one annotation for every picture. The BLEU and METEOR matrices were used for the evaluation.

In their work [15], Momin *et al.* utilized InceptionV3 and GRU, and prepared it on their BNATURE dataset, which comprised 8000 pictures (with five captions each). The model could produce good captions. To evaluate, the BLEU and METEOR were employed. Afterward, Khan *et al.* [16] proposed a model constructed on a multimodal framework consisting of a 1D-CNN and a ResNet-50. They used the "BanglaLekhaImageCaptions"[20] dataset to train their model, and it received a BLUE-1 score of 0.572. In [17], Shah *et al.* employed a transformer model and three datasets to train and analyze their model. They investigated how the model functioned, measuring the model's success with an attention based model [18]. It obtained a BLEU-1 score of 0.621 (for the three merged datasets), with a BLEU-1 score of 0.605 for the attention model (with Xception and GRU). Finally, the work [19], Humaira *et al.* examined a hybrid model consisting of two embedding layers and two separate models (BiLSTM, BiGRU) for decoding, demonstrated encouraging results.

Moreover, Xu *et al.* [3] provided the first attention based model for image captioning, which was tried on three datasets. The model identifies words for particular tiny areas of the image. The traditional RNN framework(encoder-decoder) was compared to their model, where they used "Bahdanau"[22] attention over Bi-RNNs and looked at how "soft" and "hard" attention methods performed.

This research gives an attention-model for Bengali captioning system, structured by transfer learning and a bidirectional GRU with attention (Bahdanau). When compared to current approaches, the model greatly improved performance.

III. DATA PRESENTATION

A. Dataset

To build the models, we used the "BanglaLekhaImageCaptions" dataset [20]. The collection contains 9,154 photographs as well as two Bengali annotations (captions). PNG files are used to store images. All photographs are captioned by local Bengali speakers and are relevant to the cultural and geographical setting of Bangladesh. We picked one caption for each photographs. Moreover, before utilizing the dataset to train our models, we carefully checked & corrected the dataset.

B. Data Preprocessing

We preprocessed our caption dataset to train up the models. Each caption was given a <start> and <end> token, and each word was separated to prepare it for tokenization. This provided us with a lexicon of all the data's unique words. The vocabulary was 2532 words long. We also measured the maximum caption length, which was 31 words. The word-to-index and index-to-word sequence mappings were then built. Finally, we created the caption vector by extending the sequences to the equal length as the longest one (that was 31).

IV. FEATURE EXTRACTION (ENCODER)

CNNs as the encoder are employed in captioning models to receive the images' characteristics. In order to extract visual features, state-of-the-art CNNs use a variety of methodologies and capabilities. As a result, we tested with pre-trained VGG19, ResNet50, InceptionV3, and Xception models as encoders, removing the "top layer" during the process.

- VGG19: is a model with 16 hidden layers, three dense layers, and a final SoftMax. The input pictures have been shaped into (224,224,3) for the model. VGG19 model without the "top layer" produced features of the size (49,512).
- ResNet50: has deeper networks with 50 layers. The input shape for ResNet50 is (299,299,3). The features of size (100, 2048) is the output of ResNet50, except the "top layer".
- InceptionV3 : There are 42 layers in this model, and the input shape is (299,299,3). Without the "top layer", the output shape is (64,2048).
- Xception: has a total of 71 layers. The input shape is the same as ResNet50 and InceptionV3. Without a "top layer", the result shape is (100,2048).

Therefore, the lower-convolutional layer is used to provide the features. The output makes the depiction of the "N" set of feature vectors of the "M" set of picture regions. As an outcome, each of the feature vectors represents a portion of the image. It allows the decoder to concentrate on

a particular part by only using a subset of the entire vectors. The features are stored in the NumPy files.

V. VISUAL ATTENTION MECHANISM

The human vision system is a source of inspiration for attention strategies. The human brain requires intense attentiveness to generate a solid representation of one thing at a time in a real-world situation [25]. Likewise, the attention mechanism aims to obtain picture areas one by one, allowing the model to learn greater accuracy. Machine translation and captioning models have notably benefited from these techniques. Existing mechanisms, such as "global" and "local" attention, differ in how they are using "attention" on whole visual locations or simply specific portions [23]. The "Bahdanau"[22], also known as local attention, is often estimated using the calculations below[22][23],

$$c_t = \sum_s \alpha_{ts} \bar{h}_s \quad (1)$$

$$\alpha_{ts} = \text{softmax}(\text{score}(h_t, \bar{h}_s)) \quad (2)$$

$$\text{score}(h_t, \bar{h}_s) = v_a^T \tanh(W_1 h_t + W_2 \bar{h}_s) \quad (3)$$

where, c_t is the context vector, α_{ts} is attention weight and h_t is the decoder hidden state and $\bar{h}_s$ the encoder state, v_a^T is the scaling factor, W_1, W_2 are the weights of the dense layers.

VI. MODEL

Figure 1 outlines the model framework for generating Bengali caption from an image.

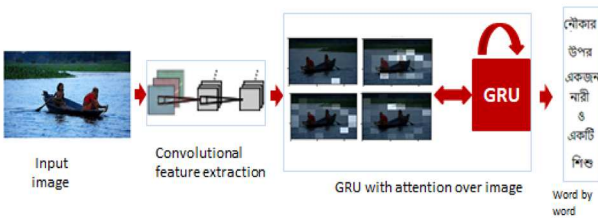


Fig. 1. Model framework for Bengali image captioning from an image.

"Bahdanau" works with GRU to form the captions. Our model receives a picture (RGB) and creates a caption (y) as a series of words where $y = \{y_1, \dots, y_c\}$, $y_i \in \mathbb{R}^k$. Here, the length of the vocabulary is k , and the range for the final reach of the words is c .

The attention model takes into account each encoder's and GRU's hidden-state to form a context vector that keeps data about certain picture regions. It aids in the selection of accurate words one by one. To setup the encoder's hidden states, the features are delivered in a dense layer, followed by a rectified linear unit (RELU). The alignment records (z_t) are achieved by relating the decoder's target hidden states, h_t (output) with each encoder's hidden states $\bar{h}_s$ (input), revealing how well the picture's s^{th} position was leveled in respect to the decoder's output of t^{th} position. It is computed by Equation (4) [23].

$$z_t = v_a^T \tanh(W_1 h_t + W_2 \bar{h}_s) \quad (4)$$

Attention weights (α_{ts}), which are known as the likelihood of key areas in the picture, are determined using the equation (5) [23],

$$\alpha_{ts} = \text{softmax}(z_t) \quad (5)$$

Afterward, the context vector (c_t) is computed by taking the mean of all image area's weights using the following formula (6) [22][23],

$$c_t = \sum_s \alpha_{ts} \bar{h}_s \quad (6)$$

Lastly, the outcomes are merged with the prior decoder state's info, which gives us the extra vector we call the last attention vector (a_t). At this point, the vector is fed through the GRU at the present step, providing our present hidden state $\tilde{h}_t$.

$$a_t = \text{Concat}(c_t, h_t), \tilde{h}_t = \text{GRU}(a_t) \quad (7)$$

Later, the present GRU hidden sequence follows two fully connected layers to provide the predictive distribution of our vocabulary-sized k target words.

$$p(y_t | y_1, \dots, y_{t-1}, x) = W_k(W_3 \tilde{h}_t) \quad (8)$$

Here, y_t is the target word, x is a given source sentence, W_k, W_3 are the weights of the dense layers.

VII. RESULT & ANALYSIS

We split the dataset into 7322 captioned pictures for training and 1832 captioned pictures for assessing the model's performance, and we ran the models for 20 epochs. Each epoch took about 35 minutes to complete (max). The models maintained the RNN 512 unit, the embedding dimension 256, and the batch size 8. Throughout the learning phase, for every batch, the decoder hidden state was set to 0, and $\langle \text{start} \rangle$ token represented the first decoder input (also applied for the evaluation part). The loss was determined using the decoder outputs. The model is end-to-end trainable and implements the target words as the decoder's input. The decoder accepts the encoder and decoder hidden states as well as the preceding predicted word in the evaluation section and produces the next predicted word. The model ends predicting by finding a $\langle \text{end} \rangle$ token. For accurate features, we tested four CNN architectures: VGG19, InceptionV3, ResNet50, and Xception, and then used our models to produce captions. Quantitative and qualitative analyses are used to evaluate the models.

The BELU (Bilingual Evaluation Understudy)[24] performance indicator was used in the quantitative analysis, and TABLE I. shows a comparison of the four encoders with the language model (decoder). The Xception's BELU scores clearly outperform the other three encoders, as shown in the table.

TABLE I. PERFORMANCE ANALYSIS

Encoder	Decoder	BLEU-1	BLEU-2	BLEU-3	BLEU-4
VGG19	Attention + GRU	80.57	78.89	75.19	71.92
InceptionV3	Attention + GRU	77.63	73.61	69.20	64.96
ResNet50	Attention + GRU	86.77	84.55	81.96	79.08
Xception	Attention + GRU	87.18	85.35	83.23	80.60

TABLE I. Quantitative analysis of the performances of different encoders (VGG19, InceptionV3, ResNet50 & Xception) using average BLEU scores for 1,2,3,4 grams.

TABLE II. further assesses the performance of our model (Xception and GRU with attention mechanism) against current state-of-the-art Bengali captioning models. The outcome indicates that our suggested model for BIC outperforms existing models with a BLUE-1 score of 87.18, a significantly improved BELU score.

TABLE II. COMPARISON

Dataset	Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4
BNATURE (their own dataset)	(DCNN+BG RU (greedy search))[15]	40.5	25.1	20.6	13.5
BANGLA LEKHAIME AGE CAPTIONS	CNN+LSTM (32 hidden layer)[13]	66.7	43.6	31.5	23.8
BANGLA LEKHAIME AGE CAPTIONS	CNN+CNN [16]	65.1	42.6	27.8	17.5
BNLIT	CNN+LSTM +BRNN (256 hidden layer)[14]	65.9	43.4	32.3	22.8
BANGLA LEKHAIME AGE CAPTIONS	Model + 2 embedding layer)[19]	67.4	52.7	45.4	34.4
BANGLA LEKHAIME AGE CAPTIONS	Transformer Model[17]	66.5	55.6	47.6	40.8
Xception (proposed model)	Attention + GRU	87.18	85.35	83.23	80.60

TABLE II. Comparison of the BLEU scores for existing works on Bengali Image Captioning.

To evaluate the model qualitatively, Figure 2 shows the caption (Bangla) generated by the model for the picture of Figure 3, where the real caption is: “*A man is walking. A train is coming from behind*” and the predicted caption is: “*A man is walking. A train is coming from behind*”. Figure 4 shows the attention parts with white rectangle boxes highlighting which parts of the image are taken into account to generate captions. Figures 5 to 16 show similar evidence to back up the proposed model's effectiveness.

BLEU Score (1-gram): 100.0
 BLEU Score (2-gram): 100.0
 BLEU Score (3-gram): 100.0
 BLEU Score (4-gram): 100.0
 Real Caption: start একজন পুরুষ হেঁটে আসছে। পিছন দিয়ে একটি রেলগাড়ি আসছে। end
 Prediction Caption: একজন পুরুষ হেঁটে আসছে। পিছন দিয়ে একটি রেলগাড়ি আসছে। end

Fig. 2. Figure 3's caption, produced by our model.



Fig. 3. Picture from the test dataset.

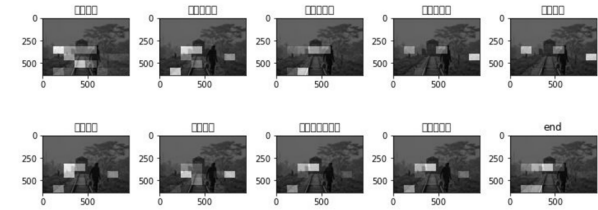


Fig. 4. White regions indicate the parts of the Fig 3 which are accountable for producing the caption.

BLEU Score (1-gram): 100.0
 BLEU Score (2-gram): 100.0
 BLEU Score (3-gram): 100.0
 BLEU Score (4-gram): 100.0
 Real Caption: start জলাশয়ের পারে অনেকগুলো নৌকা আছে। end
 Prediction Caption: জলাশয়ের পারে অনেকগুলো নৌকা আছে। end

Real Caption: There are many boats by the river.
 Prediction Caption: There are many boats by the river.

Fig. 5. Figure 6's caption, produced by our model.



Fig. 6. Picture from the test dataset.

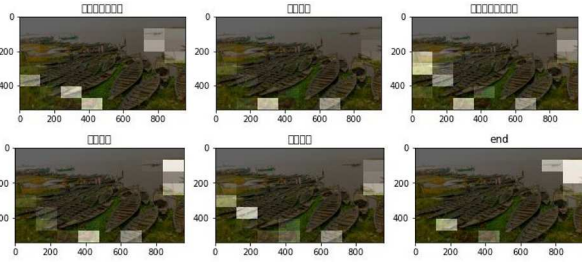


Fig. 7. White regions indicate the parts of the Fig 6 which are accountable for producing the caption.

BLEU Score (1-gram): 91.31007162822624
 BLEU Score (2-gram): 60.395935430945315
 BLEU Score (3-gram): 81.17102622306109
 BLEU Score (4-gram): 74.2614111787094
 Real Caption: start সবুজ ধান ক্ষেতের মধ্যে দিয়ে কয়েকজন ছোট ছেলে মেয়ে খেলাধুলা করছে। end
 Prediction Caption: সবুজ ধান ক্ষেতের মধ্যে দিয়ে কয়েকজন ছেলে মেয়ে খেলাধুলা করছে। end

Real Caption: A few little boys and girls are playing through the green paddy fields.

Prediction Caption: A few boys and girls are playing through the green paddy fields.

Fig. 8. Figure 9's caption, produced by our model.



Fig. 9. Picture from the test dataset.

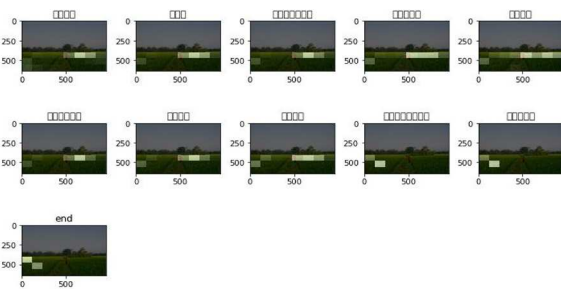


Fig. 10. White regions indicate the parts of the Fig 9 which are accountable for producing the caption.

BLEU Score (1-gram): 100.0
 BLEU Score (2-gram): 100.0
 BLEU Score (3-gram): 100.0
 BLEU Score (4-gram): 100.0
 Real Caption: start পাঞ্জাবি ও টুপি পরিহিত একজন পুরুষ রাস্তা দিয়ে হেঁটে যাচ্ছে। end
 Prediction Caption: পাঞ্জাবি ও টুপি পরিহিত একজন পুরুষ রাস্তা দিয়ে হেঁটে যাচ্ছে। end

Real Caption: A man in Punjabi and a hat walks in the road.

Prediction Caption: A man in Punjabi and a hat walks in the road.

Fig. 11. Figure 12's caption, produced by our model.



Fig. 12. Picture from the test dataset.

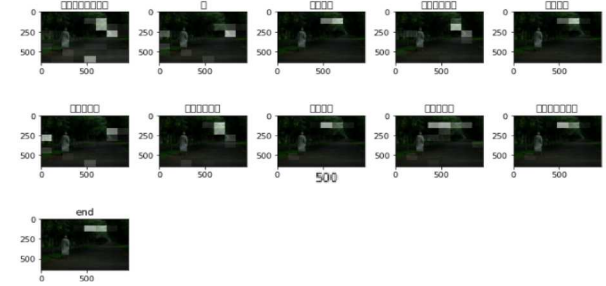


Fig. 13. White regions indicate the parts of the Fig 12 which are accountable for producing the caption.

BLEU Score (1-gram): 71.42857142857143
 BLEU Score (2-gram): 18.89822365046136
 BLEU Score (3-gram): 52.61597794341889
 BLEU Score (4-gram): 43.472087194499146
 Real Caption: start একটি নৌকায় দুইজন মানুষ দেখা যাচ্ছে। end
 Prediction Caption: একটি নৌকায় দুইজন মানুষ বসে আছে। end

Real Caption: Two people are seen in a boat.

Prediction Caption: There are two people sitting in a boat.

Fig. 14. Figure 15's caption, produced by our model.



Fig. 15. Picture from the test dataset.

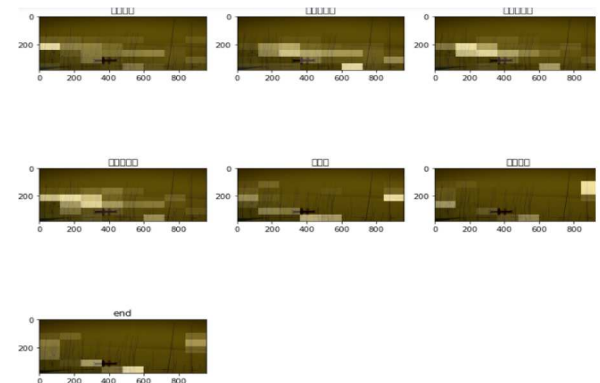


Fig. 16. White regions indicate the parts of the Fig 15 which are accountable for producing the caption.

This paper offers two considerable successes-(1) It demonstrates the efficiency of applying attention approach. (2) The outcomes of four distinct models created using various CNNs (and specifically trained on Bengali dataset) in contrast to earlier published studies. It's shown via the illustrations above that our model is capable of delivering appropriate photo captions. In Fig. 2,5,11, the captions were identical to the original. However, in Fig. 8, it was somewhat changed. In certain instances, the model's caption is nicer than the original (Fig 14). In terms of BLEU scores, all of the models' accuracy scores are far higher than earlier, and TABLE II. shows that the performance exceeds past findings.

VIII.CONCLUSION

In this paper, we propose an attention-based strategy for generating Bengali captions for Bangladeshi images. The model scans tiny picture areas and builds descriptions in a natural language one word at a time. We investigated caption quality using the VGG19, Inceptionv3, Resnet50, and Xception architectures as encoders and GRU with an attention approach as a decoder. Our models based on different encoders yield promising outcomes. Moreover, our Xception-based-model appears to be better than prior attempts. This is why, in this research, we picked our Xception-based-model as our preferred attention-model. Then we showed various examples of our model's caption prediction for Bangladesh's geographical settings and Bengali language. We assessed the model's results to those of other Bengali models as well. We aim to expand the dataset in the future and train the model with more annotations for each picture. The global attention process may also be investigated for Bengali captioning models.

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